HW4

Yi-Ting Tsai

Problem 1 - Tidyverse: New Zealand

Import library and read file

```
library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.5.3
```

```
Warning: package 'ggplot2' was built under R version 4.5.3
```

```
Warning: package 'tibble' was built under R version 4.5.3
```

```
Warning: package 'tidyr' was built under R version 4.5.3
```

```
Warning: package 'readr' was built under R version 4.5.3
```

```
Warning: package 'purrr' was built under R version 4.5.3
```

```
Warning: package 'dplyr' was built under R version 4.5.3
```

```
Warning: package 'stringr' was built under R version 4.5.3
```

```
Warning: package 'forcats' was built under R version 4.5.3
```

```
Warning: package 'lubridate' was built under R version 4.5.3
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```r
nzge <- readRDS("nzge.rds")
```

a. Generate a table

```r
nzge %>%
  group_by(election_year, voting_type) %>%
  summarise(
    total_votes = sum(votes),
    .groups = "drop"
  ) %>%
  arrange(desc(total_votes))
```

```
# A tibble: 10 x 3
   election_year voting_type total_votes
           <dbl> <chr>             <dbl>
 1          2014 Party           2416479
 2          2014 Candidate       2375493
 3          2008 Party           2356536
 4          2008 Candidate       2325598
 5          2005 Party           2286190
 6          2005 Candidate       2260670
 7          2011 Party           2257336
 8          2011 Candidate       2225766
 9          2002 Party           2040248
10          2002 Candidate       2022115
```

b. Report the proportion of votes

```r
nzge %>%
  filter(
    election_year == "2014",
    voting_type == "Candidate"
  ) %>%
  group_by(party) %>%
  summarise(
    candidate_vote = sum(votes),
    .groups = "drop"
  ) %>%
  mutate(
    propartion = candidate_vote / sum(candidate_vote)
  ) %>%
  arrange(desc(propartion)) %>%
  mutate(
    percentage = sprintf("%.2f%%", propartion * 100)
  ) %>%
  select(party, candidate_vote, percentage)
```

```
# A tibble: 25 x 3
   party                     candidate_vote percentage
   <chr>                              <dbl> <chr>
 1 National Party                   1081787 45.54%
 2 Labour Party                      801287 33.73%
 3 Green Party                       165718 6.98%
 4 Conservative Party                 81075 3.41%
 5 New Zealand First Party            73384 3.09%
 6 Maori Party                        42108 1.77%
 7 MANA Movement                      32333 1.36%
 8 Informal Candidate Votes           27886 1.17%
 9 ACT New Zealand                    27778 1.17%
10 United Future                      14722 0.62%
# i 15 more rows
```

c. Which party won the Candidate vote and which party won the Party vote

```
nzge %>%
  group_by(election_year, voting_type, party) %>%
  summarise(
    total_votes = sum(votes),
    .groups = "drop"
  ) %>%
  group_by(election_year, voting_type) %>%
  slice_max(
    order_by = total_votes,
    n = 1,
    with_ties = TRUE
  ) %>%
  summarise(
    winner = paste(sort(party), collapse = ": "),
    .groups = "drop"
  )
```

```
# A tibble: 10 x 3
   election_year voting_type winner
           <dbl> <chr>       <chr>
 1          2002 Candidate   Labour Party
 2          2002 Party       Labour Party
 3          2005 Candidate   National Party
 4          2005 Party       Labour Party
 5          2008 Candidate   National Party
 6          2008 Party       National Party
 7          2011 Candidate   National Party
 8          2011 Party       National Party
 9          2014 Candidate   National Party
10          2014 Party       National Party
```

Problem 2 - Tidyverse: Tennis

```
library(tidyverse)
tennis <- read.csv("tennis-atp-sackmann-data-master/atp_matches_2019.csv")
```

a. How many tournaments

```
tennis %>%
  summarise(
    tournaments = n_distinct(tourney_name)
  )
```

```
  tournaments
1         128
```

b. More than one tournament

```
tennis %>%
  filter(
    round == "F"
  ) %>%
  group_by(winner_name) %>%
  count(winner_name, name = "tournaments_won") %>%
  filter(
    tournaments_won > 1
  ) %>%
  arrange(desc(tournaments_won))
```

```
# A tibble: 12 x 2
# Groups:   winner_name [12]
   winner_name        tournaments_won
   <chr>                        <int>
 1 Dominic Thiem                    5
 2 Novak Djokovic                   5
 3 Daniil Medvedev                  4
 4 Rafael Nadal                     4
 5 Roger Federer                    4
 6 Alex De Minaur                   3
 7 Stefanos Tsitsipas               3
 8 Benoit Paire                     2
 9 Cristian Garin                   2
10 Jo-Wilfried Tsonga               2
11 Matteo Berrettini                2
12 Nick Kyrgios                     2
```

12 players won more than 1 tournaments in 2019. Dominic Thiem and Novak Djokovic won the most, 5 tournaments each.

c. Winners have more aces than losers?

```
tennis %>%
  filter(!is.na(w_ace), !is.na(l_ace)) %>%
  mutate(ace_diff = w_ace - l_ace) %>%
  summarise(
    matches = n(),
    mean_winner_aces = mean(w_ace),
    mean_loser_aces = mean(l_ace),
    mean_difference = mean(ace_diff),
    median_difference = median(ace_diff),
    proportion_winner_more = mean(ace_diff > 0),
    proportion_equal = mean(ace_diff == 0),
    proportion_loser_more = mean(ace_diff < 0)
  )
```

```
  matches mean_winner_aces mean_loser_aces mean_difference median_difference
1    2694         7.497402        5.792502          1.7049                 1
  proportion_winner_more proportion_equal proportion_loser_more
1              0.5727543       0.08351893             0.3437268
```

These descriptive results support the claim that winners had more aces on average in the observed matches.

d. Highest win-rate

```
full_join(
  tennis %>% count(winner_name, name = "wins"),
  tennis %>% count(loser_name, name = "losses"),
  by = c("winner_name" = "loser_name")
) %>%
  mutate(
    wins = replace_na(wins, 0L),
    losses = replace_na(losses, 0L),
```

```
    matches = wins + losses,
    win_rate = wins / matches
  ) %>%
  filter(matches >= 5) %>%
  slice_max(win_rate, n = 1, with_ties = TRUE)
```

```
   winner_name wins losses matches  win_rate
1 Rafael Nadal   60      9      69 0.8695652
```

Problem 3 - Visualization

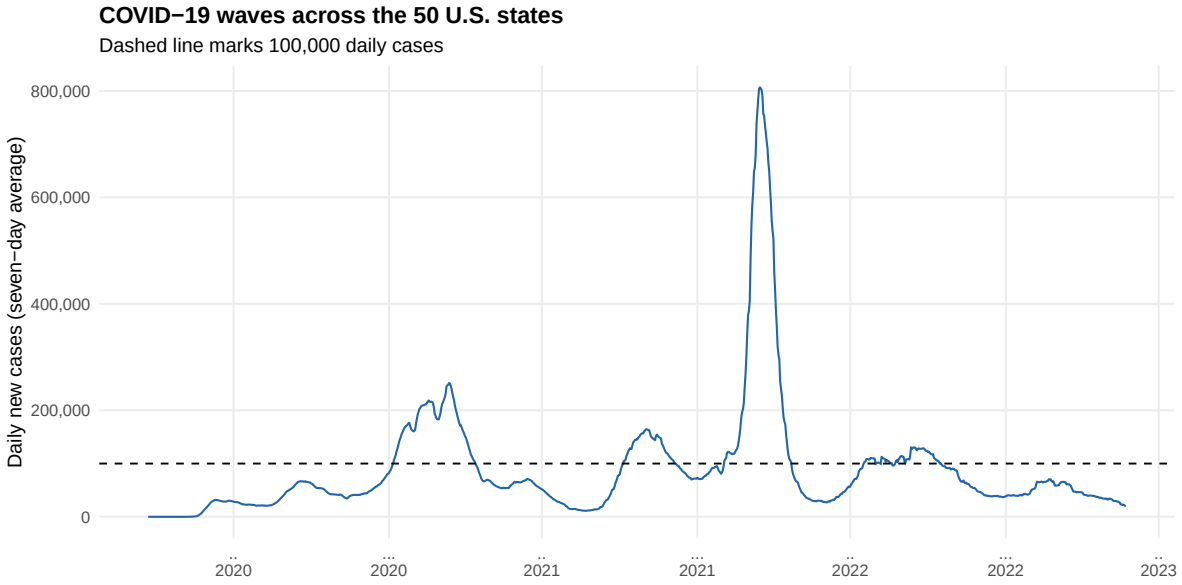
```
library(tidyverse)
covid <- read.csv("https://raw.githubusercontent.com/nytimes/covid-19-data/refs/heads/master/

theme_set(
  theme_minimal(base_size = 12) +
    theme(
      plot.title = element_text(face = "bold"),
      panel.grid.minor = element_blank(),
      plot.caption = element_text(hjust = 0)
    )
)
```

a. How many major and minor spikes in cases were there?

```
covid %>%
  mutate(date = as.Date(date)) %>%
  group_by(date) %>%
  summarise(
    cases_avg = sum(cases_avg),
    .groups = "drop"
  ) %>%
  ggplot(aes(date, cases_avg)) +
  geom_line(color = "#2166AC", linewidth = 0.6) +
  geom_hline(yintercept = 100000, linetype = "dashed") +
  scale_y_continuous(labels = scales::label_comma()) +
  scale_x_date(date_breaks = "6 months", date_labels = "%b\n%Y") +
```

```
  labs(
    title = "COVID-19 waves across the 50 U.S. states",
    subtitle = "Dashed line marks 100,000 daily cases",
    x = NULL,
    y = "Daily new cases (seven-day average)"
  )
```



I counted 4 major spikes and 4 minor spikes. I treated distinct waves with peaks of at least 100,000 daily cases as major spikes, and smaller but clearly separated waves as minor spikes. Small fluctuations within a wave were not counted as separate spikes.

b. What differences do you see in their trajectories over time?

```
state_rates <- covid %>%
  filter(state %in% state.name) %>%
  group_by(state) %>%
  summarise(
    overall_rate = sum(cases_avg_per_100k),
    .groups = "drop"
  )

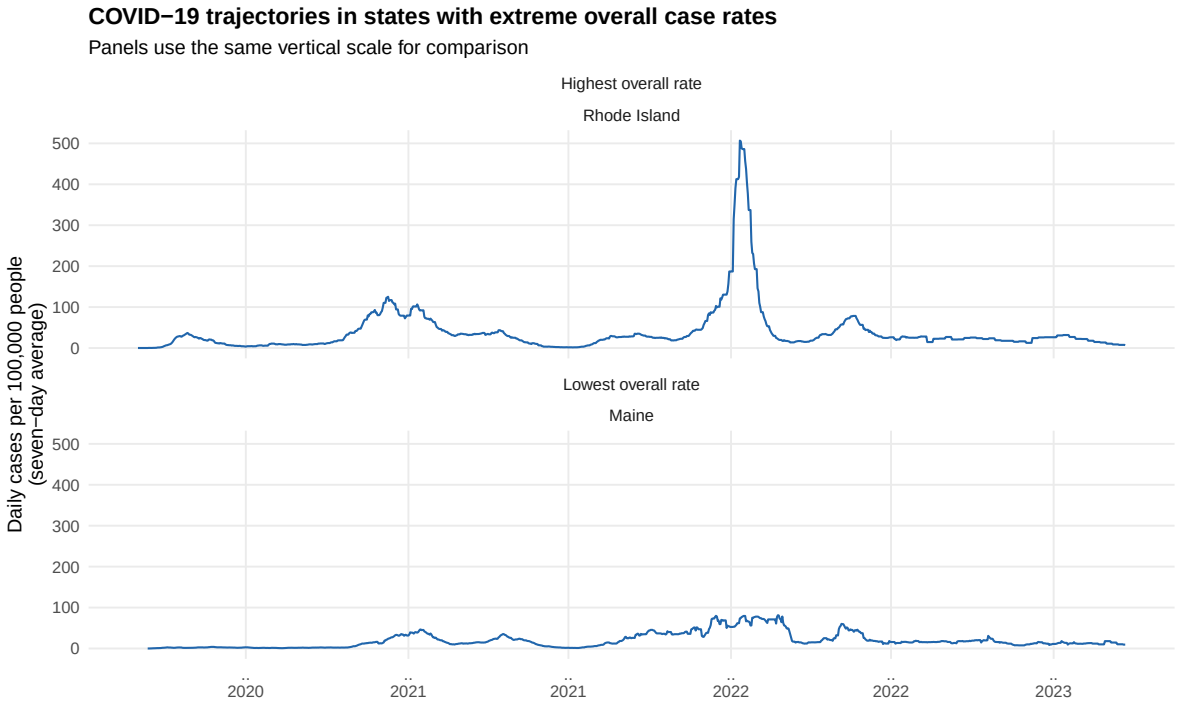
highest_state <- state_rates %>%
```

```r
  slice_max(overall_rate, n = 1, with_ties = TRUE)

lowest_state <- state_rates %>%
  slice_min(overall_rate, n = 1, with_ties = TRUE)

selected_states <- bind_rows(
  highest_state %>% mutate(group = "Highest overall rate"),
  lowest_state %>% mutate(group = "Lowest overall rate")
)

covid %>%
  mutate(date = as.Date(date)) %>%
  inner_join(selected_states, by = "state") %>%
  ggplot(aes(x = date, y = cases_avg_per_100k)) +
  geom_line(linewidth = 0.6, color = "#2166AC") +
  facet_wrap(vars(group, state), ncol = 1) +
  scale_x_date(date_breaks = "6 months", date_labels = "%b\n%Y") +
  labs(
    title = "COVID-19 trajectories in states with extreme overall case rates",
    subtitle = "Panels use the same vertical scale for comparison",
    x = NULL,
    y = "Daily cases per 100,000 people\n(seven-day average)"
  )
```

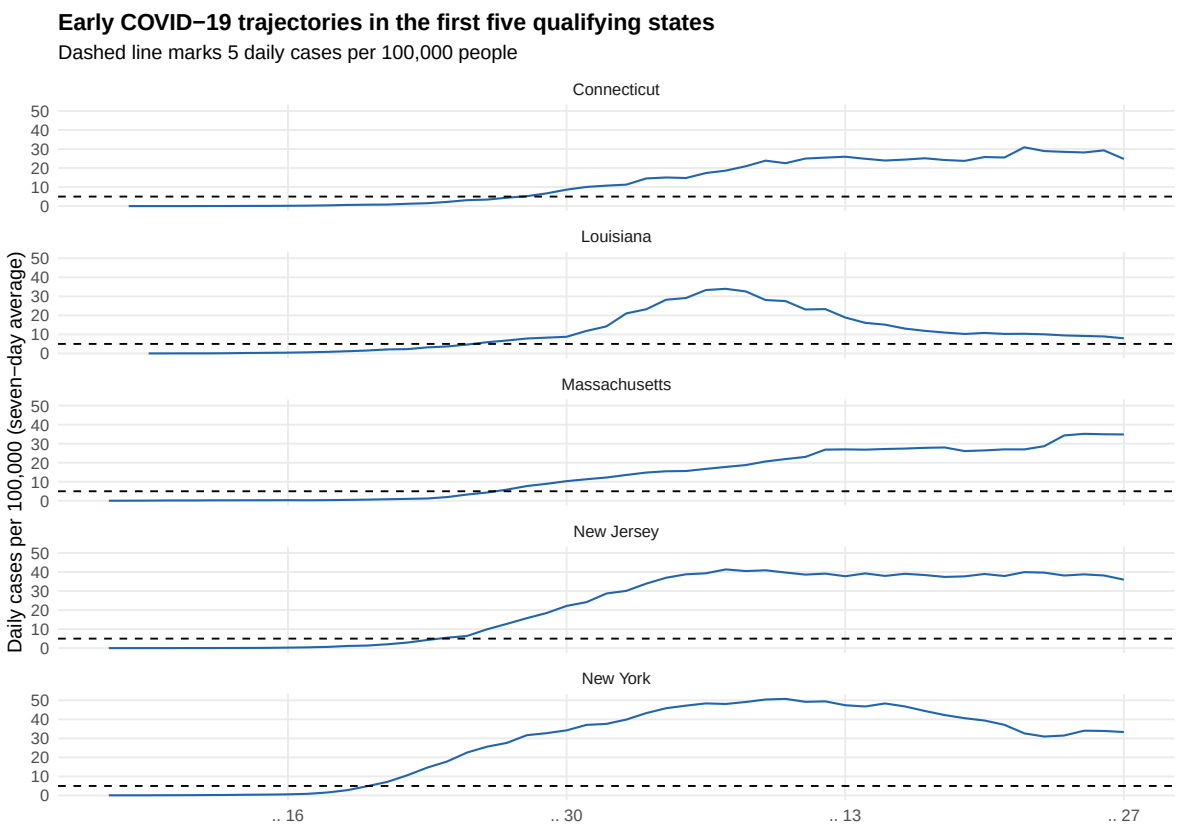


Rhode Island had the highest overall reported case rate, while Maine had the lowest. The largest contrast occurred in early 2022: Rhode Island reached approximately 500 daily cases per 100,000 people, compared with about 80 in Maine. Rhode Island's peak was sharp, while Maine experienced a broader period of moderately elevated case rates. Overall, the plot suggests that Rhode Island's higher cumulative reported case rate reflected larger waves, especially the sharp surge in early 2022, rather than consistently higher rates at every point in time.

c. The first five states to experience Covid in a substantial way

```
first_five <- covid %>%
  mutate(date = as.Date(date)) %>%
  filter(cases_avg_per_100k >= 5) %>%
  group_by(state) %>%
  summarise(
    first_date = min(date),
    .groups = "drop"
  ) %>%
  arrange(first_date, state) %>%
  slice_head(n = 5)
```

```r
covid %>%
  mutate(date = as.Date(date)) %>%
  filter(
    state %in% first_five$state,
    date >= min(first_five$first_date) - 14,
    date <= max(first_five$first_date) + 30
  ) %>%
  ggplot(aes(date, cases_avg_per_100k)) +
  geom_line(color = "#2166AC", linewidth = 0.6) +
  geom_hline(yintercept = 5, linetype = "dashed") +
  facet_wrap(vars(state), ncol = 1) +
  scale_x_date(date_breaks = "2 weeks", date_labels = "%b %d") +
  labs(
    title = "Early COVID-19 trajectories in the first five qualifying states",
    subtitle = "Dashed line marks 5 daily cases per 100,000 people",
    x = NULL,
    y = "Daily cases per 100,000 (seven-day average)"
  )
```



I defined substantial impact as reaching a seven-day-average daily case rate of at least 5 cases per 100,000 people. Under this definition, the first five states were Connecticut, Louisiana, Massachusetts, New Jersey, New York. The plots show when these states reached the threshold. This ranking depends on the chosen threshold and reflects reported cases, not the true start of transmission.